Axial and Appendicular Skeleton

The adult skeleton has 206 bones. Anatomists sort them into two divisions. The **axial skeleton** forms the long axis of the body and protects the brain, spinal cord and organs of the chest. The **appendicular skeleton** carries the limbs and the girdles that attach them to the axial skeleton.



Figure 1. Skull, vertebral column, rib cage and upper limbs.

Axial skeleton (80 bones)

Region	Bones	Main role
Skull	22 cranial and facial bones	Protects the brain; holds the eyes, ears and teeth
Middle ear	6 ossicles	Carry sound to the inner ear
Hyoid	1	Anchors the tongue
Vertebral column	26 (24 vertebrae, sacrum, coccyx)	Supports the head and trunk; shields the spinal cord
Thoracic cage	24 ribs and the sternum	Protects the heart and lungs; helps breathing

Appendicular skeleton (126 bones)

Region	Bones	Main role
Pectoral girdle	Clavicles and scapulae (4)	Attach the arms to the trunk
Upper limbs	Humerus, radius, ulna, carpals, metacarpals, phalanges (60)	Reach, grip and fine movement
Pelvic girdle	Two hip bones (2)	Carry body weight to the legs
Lower limbs	Femur, patella, tibia, fibula, tarsals, metatarsals, phalanges (60)	Stand, walk and run

> **Remember:** 80 axial + 126 appendicular = 206 bones. Babies are born with about 270 bones; many fuse as they grow.

Check yourself

1. Which division contains the sternum?
2. Name the two bones of the pectoral girdle.
3. Why is the pelvic girdle heavier than the pectoral girdle?

Anthropotomy demo handout. Source: OpenStax Anatomy and Physiology 2e, chapter 7 (CC BY 4.0).